

Radar Search Pipeline TO BE 0.7.6.3.6.11

Review design for a planned Radar search pipeline change

Status: TO BE

Slice: 0.7.6.3.6.11: Completion target prioritization for uncovered benchmark targets

Product area: Radar candidate and signal search

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1. Decision Context

The expanded benchmark_smoke_plus run proved that a larger bounded budget materially improves SIBUR contour recall:

- strict recall reached 1.0;
- review recall stayed at 0.6667;
- the only missed baseline entity was tobolsk-site.

The remaining miss is no longer a hidden budget problem. Radar generated an executable expansion target for Tobolsk production site, but the selection layer ranked incidental production-site targets ahead of that explicit benchmark target.

2. AS IS Problem

Current target selection sorts completion targets mostly by target lane, numeric priority, and query text. It does not distinguish:

- explicit benchmark/baseline targets;
- source-backed incidental targets found during retrieval;
- generic industrial search terms;
- noisy document-like labels;
- numeric-only labels.

As a result, a known uncovered benchmark target can lose completion slots to less important targets such as document titles, generic site terms, or incidental branches.

3. Intended Behavior

The selector must rank targets by intent before ranking by text.

Priority order inside a lane:

1. explicit benchmark/baseline targets that are not already selected;
2. concrete named targets from retrieved source evidence;
3. generic source-backed universe gaps;
4. noisy document-like labels;
5. numeric-only labels.

This priority must apply to:

- guaranteed lane selection;
- bounded completion selection;
- optional variant fill after guarantees and completion.

The scheduler remains the admission owner. The selector only decides what work should be offered to the scheduler first.

4. Roles Changed

Role	Change
Search expansion service	Adds target origin metadata to expansion targets.
Search expansion selector	Uses target origin and label quality to rank guaranteed and completion targets.
Dossier/report projection	Shows why a target was prioritized or deprioritized.
Evaluation	Keeps completion_not_selected diagnostics, now with better selection reasons.

5. Context Passed Between Roles

Expansion targets receive additive fields:

- target_origin;
- completion_rank;
- completion_rank_reason;
- deprioritized_reason;
- uncovered_baseline_target.

These fields are product-safe. They contain no prompts, provider dumps, headers, credentials, or hidden reasoning.

6. Selection Flow

<!-- diagram: completion target prioritization -->

7. Ranking Rules

The selector computes a stable rank key:

1. lane priority;
2. target origin priority;
3. label quality penalty;
4. target priority;
5. query text.

Origin priority:

Origin	Priority
benchmark_context	0
retrieved_source	10
candidate_gap	20
generated_alias	30

Origin	Priority
unknown	40

Label penalties:

Label pattern	Penalty
clean concrete named target	0
generic industrial phrase	20
document-like prefix such as pdf	30
numeric-only label	40

8. Budget Semantics

No budget is bypassed.

The selector may move a target earlier, but the work scheduler and existing external-call guards still decide whether it can run.

If a high-priority target cannot run, the dossier must say whether it was:

- not executable;
- not admitted by scheduler;
- blocked by source policy;
- blocked by budget;
- searched without support;
- projected incorrectly.

9. Test Plan

Unit tests must verify the changed selector directly:

- explicit benchmark production-site target outranks incidental retrieved production-site targets;
- clean named target outranks document-like pdf ... target;
- clean named target outranks numeric-only target;
- completion_target_limit=5 selects a tobolsk-site-like benchmark target before incidental production-site targets;
- ranking diagnostics explain why skipped targets lost.

Report/evaluation tests must verify:

- target origin and rank reasons are visible in dossier/report payloads;
- a missed explicit target is no longer explained only as generic completion_not_selected when it was deprioritized incorrectly.

10. Acceptance

Fast acceptance:

- selector tests prove benchmark targets are selected before incidental targets;
- benchmark report tests expose ranking metadata;
- existing scheduler/admission tests remain green.

Manual acceptance:

- run bounded benchmark_smoke_plus;
- tobolsk-site must either be selected/executed or receive a more precise blocker than plain completion_not_selected.

11. Out Of Scope

- No scoring change.
- No UI change.
- No new provider integration.
- No SIBUR hardcode in production runtime.
- No automatic move to benchmark_live.